Department of Mechanical and Production Engineering

Aarhus University

Exercises

Author	Noah Rahbek Bigum Hansen	202405538
Course	Varmetransmission	290221U002
Instructor	Pourya Forooghi & Mugisa Y. Kano	
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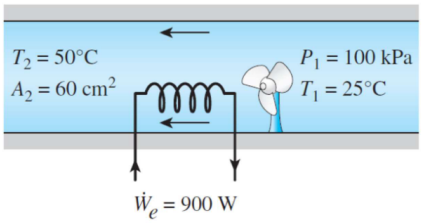
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Lecture 1: Introduction to Heat Transfer 2026-08-25

Problem 1–31

A hair dryer is basically a duct in which a few layers of electric resistors are placed. A small fan pulls the air in and forces it to flow over the resistors where it is heated. Air enters a 900 W hair dryer at 100 kPa and 25 °C, and leaves at 50 °C.
The cross-sectional area of the hair dryer at the exit is 60 cm^2. Neglecting the power consumed by the fan and the heat losses through the walls of the hair dryer, determine



(a) the volume flow rate of air at the inlet and

(b) the velocity of the air at the exit.

Givens

$\dot{W}_e = 900\,\text{W}$	Power consumption of the hair dryer
$P_1 = 100\,\text{kPa}$	Pressure at the inlet
$T_1 = 25\,^\circ\text{C}$	Temperature at the inlet
$T_2 = 50\,^\circ\text{C}$	Temperature of the outlet
$A_2 = 60\,\text{cm}^2$	Area of the outlet

Assumptions

- The air is an ideal gas
- The system is at steady state, $\frac{\mathrm{d}}{\mathrm{d}t} = 0$
- The fan does not consume power and no heat is lost through the wall
- c_p is assumed constant such that the enthalpy change is simply $c_p(T_2 - T_1)$
- The change in pressure and potential and kinetic energy of the air is negligible

Derivation

(a) Using the ideal gas law

$$p = \rho RT$$

we can solve for the density of the air at the inlet and outlet respectively, remembering we assume the pressure is the same at both places

$$\rho = \frac{p}{RT}$$
$$\rho_1 = \frac{100\,\text{kPa}}{298.15\,\text{K} \cdot 0.287\,\text{kPa}\,\text{m}^3/\text{kg}\,\text{K}} = 1.169\,\text{kg/m}^3$$
$$\rho_2 = \frac{100\,\text{kPa}}{323.15\,\text{K} \cdot 0.287\,\text{kPa}\,\text{m}^3/\text{kg}\,\text{K}} = 1.078\,\text{kg/m}^3.$$

As the entirety of the heat supplied by the heating element is assumed to go into the air stream the energy balance becomes

$$\dot{W}_e = \dot{m} c_p (T_2 - T_1) = \rho_1 Q_V c_p (T_2 - T_1).$$

We can now solve for the volume flow rate, Q_V as

$$Q_V = \frac{\dot{W}_e}{(T_2 - T_1) c_p \cdot \rho_1} = \frac{900\,\text{W}}{25\,\text{K} \cdot 1007\,\text{J}\,\text{kg}^{-1}\,\text{K}^{-1} \cdot 1.169\,\text{kg}\,\text{m}^{-3}} = 0.030\,58\,\text{m}^3\,\text{s}^{-1}.$$

> **Result**
>
> The volume flow rate through the hair dryer is
>
> $$Q_V = 0.031\,\text{m}^3\,\text{s}^{-1}.$$

(b) We can use a similar formula as above, however, this time dividing by the outlet area as well to obtain the outlet flow speed:

$$V_2 = \frac{\dot{W}_e}{(T_2 - T_1) c_p \cdot \rho_2 \cdot A_2} = \frac{900\,\text{W}}{25\,\text{K} \cdot 1007\,\text{J}\,\text{kg}^{-1}\,\text{K}^{-1} \cdot 1.078\,\text{kg}\,\text{m}^{-3} \cdot 60\,\text{cm}^2} = 5.526\,\text{m}\,\text{s}^{-1}.$$

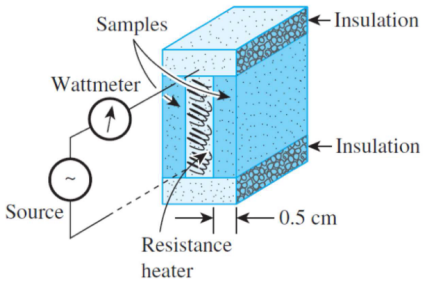
> **Result**
>
> The outlet flow speed of the air is
>
> $$V_2 = 5.526\,\text{m}\,\text{s}^{-1}.$$

> **Final result – Problem 1**
>
> **(a)** The volume flow rate through the hair dryer is
>
> $$Q_V = 0.031\,\text{m}^3\,\text{s}^{-1}.$$
>
> **(b)** The outlet flow speed of the air is
>
> $$V_2 = 5.526\,\text{m}\,\text{s}^{-1}.$$

Problem 1–59

One way of measuring the thermal conductivity of a material is to sandwich an electric thermofoil heater between two identical rectangular samples of the material and to heavily insulate the four outer edges, as shown in the figure. Thermocouples attached to the inner and outer surfaces of the samples record the temperatures.

During an experiment, two 0.5 cm thick samples 10 cm × 10 cm in size are used. When steady operation is reached, the heater is observed to draw 25 W of electric power, and the temperature of each sample is observed to drop from 82 °C at the inner surface to 74 °C at the outer surface. Determine the thermal conductivity of the material at the average temperature.



Givens

$h = 0.5\,\text{cm}$	Height of specimen
$w = 10\,\text{cm}$	Width of specimen
$\dot{Q}_{\text{heater}} = 25\,\text{W}$	Power-consumption of heater
$\Delta T = 8\,^{\circ}\text{C}$	Temperature drop across specimen

Assumptions

- Perfectly insulated outer edges

Derivation

As the heater is always heating two samples at a time the power of the heater is split evenly between the two samples:

$$\dot{Q}_{\text{sample}} = \frac{\dot{Q}_{\text{heater}}}{2} = 12.5\,\text{W}.$$

This flow of $\dot{Q}_{\text{sample}} = 12.5\,\text{W}$ is what sustains the temperature of the samples. The thermal conductivity of a material is defined from:

$$\dot{Q}_{\text{cond}} = -kA\frac{\Delta T}{\Delta x}.$$

Hence,

$$k = \frac{\dot{Q}_{\text{cond}} \cdot h}{A \cdot \Delta T} = \frac{12.5\,\text{W} \cdot 0.5\,\text{cm}}{10\,\text{cm} \cdot 10\,\text{cm} \cdot 8\,\text{K}} = 0.78\,\text{W}\,\text{m}^{-1}\,\text{K}^{-1}.$$

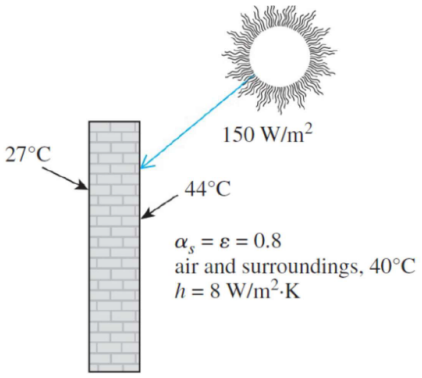
Final result – Problem 2

Hence the thermal conductivity of each sample is:

$$k = 0.78\,\text{W}\,\text{m}^{-1}\,\text{K}^{-1}.$$

Problem 1–100

The inner and outer surfaces of a 25 cm thick wall in summer are at 27 °C and 44 °C, respectively. The outer surface of the wall exchanges heat by radiation with surrounding surfaces at 40 °C, and convection heat transfer coefficient of $8\,\mathrm{W\,m^{-2}\,K^{-1}}$. Solar radiation is incident on the surface at a rate of $150\,\mathrm{W\,m^{-2}}$. If both the emissivity and the solar absorptivity of the outer surface are 0.8, determine the effective thermal conductivity of the wall.



Givens

$w = 25\,\mathrm{cm}$	Thickness of wall
$T_{\mathrm{inner}} = 27\,^{\circ}\mathrm{C}$	Inner temperature of wall
$T_{\mathrm{outer}} = 44\,^{\circ}\mathrm{C}$	Outer temperature of wall
$T_{\infty,\mathrm{outer}} = 40\,^{\circ}\mathrm{C}$	Ambient temperature outside wall
$h = 8\,\mathrm{W/m^2/K}$	Heat transfer coefficient
$\dot{q}_{\mathrm{indicent}} = 150\,\mathrm{W\,m^{-2}}$	Solar radiation incident on outer wall
$\epsilon = \alpha = 0.8$	Solar emissivity and absorptivity of wall

Assumptions

- The system has reached steady-state

Derivation

The radiation heat transfer rate on a solid surface is:

$$\dot{q}_{\mathrm{rad}} = \dot{q}_{\mathrm{emit}} - \dot{q}_{\mathrm{abs}}.$$

Here, the net emitted radiation can be found using Boltzmann's law:

$$\dot{q}_{\mathrm{emit}} = \epsilon\sigma\left(T_s^4 - T_\infty^4\right) = 0.8 \cdot 5.670 \cdot 10^{-8}\,\mathrm{W\,m^{-2}\,K^{-4}} \cdot \left((44\,^{\circ}\mathrm{C})^4 - (40\,^{\circ}\mathrm{C})^4\right) = 22.7\,\mathrm{W\,m^{-2}}.$$

The absorbed radiation is found as:

$$\dot{q}_{\mathrm{abs}} = \alpha \cdot \dot{q}_{\mathrm{incident}} = 0.8 \cdot 150\,\mathrm{W\,m^{-2}} = 120\,\mathrm{W\,m^{-2}}.$$

Hence, the radiation heat transfer rate on the outside of the wall is:

$$\dot{q}_{\text{rad}} = 22.7\,\text{W}\,\text{m}^{-2} - 120\,\text{W}\,\text{m}^{-2} = 97.3\,\text{W}\,\text{m}^{-2}.$$

From Newton's law of cooling, the convection heat transfer rate is:

$$\dot{q}_{\text{conv}} = h\,(T_{\text{outer}} - T_{\infty}) = 8\,\text{W}\,\text{m}^{-2}\,\text{K}^{-1} \cdot (44\,\text{K} - 40\,\text{K}) = 31\,\text{W}\,\text{m}^{-2}.$$

Hence, the outside of the wall is gaining heat at a rate of:

$$\dot{q}_{\text{outer}} = \dot{q}_{\text{rad}} - \dot{q}_{\text{conv}} = 97.3\,\text{W}\,\text{m}^{-2} - 31\,\text{W}\,\text{m}^{-2} = 66.3\,\text{W}\,\text{m}^{-2}.$$

Now, the thermal conductivity is easily found as:

$$k = \frac{\dot{q}_{\text{outer}} \cdot w}{T_{\text{outer}} - T_{\text{inner}}} = \frac{66.3\,\text{W}\,\text{m}^{-2} \cdot 25\,\text{cm}}{44\,^{\circ}\text{C} - 27\,^{\circ}\text{C}} = 0.975\,\text{W}\,\text{m}^{-1}\,\text{K}^{-1}.$$

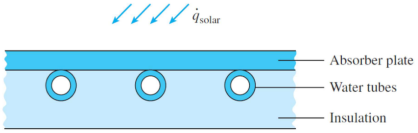
Final result – Problem 3

Hence, the thermal conductivity of the wall is

$$k = 0.975\,\text{W}\,\text{m}^{-1}\,\text{K}^{-1}.$$

Problem 1–106

A flat-plate solar collector is used to heat water by having water flow through tubes attached at the back of the thin solar absorber plate. The absorber plate has a surface area of $2\,\mathrm{m}^2$ with emissivity and absorptivity of 0.9. The surface temperature of the absorber is $35\,^\circ\mathrm{C}$, and solar radiation is incident on the absorber at $500\,\mathrm{W\,m^{-2}}$ with a surrounding temperature of $0\,^\circ\mathrm{C}$. Convection heat transfer coefficient at the absorber surface is $5\,\mathrm{W\,m^{-2}\,K^{-1}}$, while the ambient temperature is $25\,^\circ\mathrm{C}$. Net heat rate absorbed by the solar collector heats the water from an inlet temperature (T_{in}) to an outlet temperature (T_{out}). If the water flow rate is $5\,\mathrm{g\,s^{-1}}$ with specific heat of $4.2\,\mathrm{kJ\,kg^{-1}\,K^{-1}}$, determine the temperature rise of the water.



Assumptions

- The system has reached steady-state

Givens

$A_s = 2\,\mathrm{m}^2$	Surface area of absorber
$\epsilon = \alpha = 0.9$	Emissivity and absorptivity of absorber
$T_{\mathrm{absorber}} = 35\,^\circ\mathrm{C}$	Temperature of absorber
$\dot{q}_{\mathrm{incident}} = 500\,\mathrm{W\,m^{-2}}$	Solar radiation incident at absorber
$T_{\mathrm{rad},\infty} = 0\,^\circ\mathrm{C}$	Effective surrounding temperature for radiation
$h = 5\,\mathrm{W\,m^{-2}\,K^{-1}}$	Heat transfer coefficient
$T_{\mathrm{conv.},\infty} = 25\,^\circ\mathrm{C}$	Ambient temperature for convection
$\dot{m} = 5\,\mathrm{g\,s^{-1}}$	Mass flow rate of water
$c_p = 4.2\,\mathrm{kJ\,kg^{-1}\,K^{-1}}$	Specific heat capacity of water.

Derivation

For radiation we use the same procedure as above and get:

$$\dot{q}_{\mathrm{rad}} = \epsilon\sigma\left(T_{\mathrm{absorber}}^4 - T_{\mathrm{rad},\infty}^4\right) = 0.9 \cdot 5.670 \cdot 10^{-8}\,\mathrm{W\,m^{-2}\,K^{-4}} \cdot \left((35\,^\circ\mathrm{C})^4 - (0\,^\circ\mathrm{C})^4\right) = 176.05\,\mathrm{W\,m^{-2}}.$$

For convection, we also use the same procedure as above, which yields:

$$\dot{q}_{\text{conv}} = h\left(T_{\text{absorber}} - T_{\text{conv.},\infty}\right) = 5\,\text{W}\,\text{m}^{-2}\,\text{K}^{-1} \cdot (35\,^{\circ}\text{C} - 25\,^{\circ}\text{C}) = 50\,\text{W}\,\text{m}^{-2}.$$

Hence, the total energy balance is:

$$\dot{q}_{\text{absorber}} = \dot{q}_{\text{rad}} - \dot{q}_{\text{conv}} = 176.05\,\text{W}\,\text{m}^{-2} - 50\,\text{W}\,\text{m}^{-2} = 126.05\,\text{W}\,\text{m}^{-2}.$$

This means the total absorbed power over the entire area of the absorber is:

$$\dot{Q}_{\text{absorber}} = \dot{q}_{\text{absorber}} \cdot A_s = 126.05\,\text{W}\,\text{m}^{-2} \cdot 2\,\text{m}^2 = 252.1\,\text{W}.$$

As the system is assumed to be at steady state none of this heat will go to warming the absorber – instead all the $\dot{Q}_{\text{absorber}} = 252.1\,\text{W}$ goes towards heating the water. Using the formula for temperature change as a result of added thermal energy we arrive at:

$$\begin{aligned}
\dot{Q}_{\text{absorber}} &= \dot{m} c_p \cdot \Delta T \\
\Delta T &= \frac{\dot{Q}_{\text{absorber}}}{\dot{m} \cdot c_p} \\
&= \frac{252.1\,\text{W}}{5\,\text{g}\,\text{s}^{-1} \cdot 4.2\,\text{kJ}\,\text{kg}^{-1}\,\text{K}^{-1}} \\
&= 12\,\text{K}.
\end{aligned}$$

Final result – Problem 4

Hence the temperature of the water will increase by:

$$\Delta T = 12\,\text{K}$$

when passing through the absorber.

Lecture 2: Conduction, Pt. I

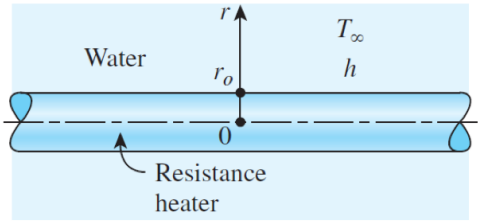
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No Exercises for this lecture :)

Lecture 3: Conduction, Pt. II 2026-09-8

Problem 2–97

A long homogeneous resistance wire of radius $r_o = 0.6\,\text{cm}$ and thermal conductivity $k = 15.2\,\text{W m}^{-1}\,\text{K}^{-1}$ is being used to boil water at atmospheric pressure by the passage of electric current. Heat is generated in the wire uniformly as a result of resistance heating at a rate of $16.4\,\text{W cm}^{-3}$. The heat generated is transferred to water at 100 °C convection with an average heat transfer coefficient of $h = 3200\,\text{W m}^{-2}\,\text{K}^{-1}$. Assuming steady one-dimensional heat transfer,



(a) express the differential equation and the boundary conditions for heat conduction through the wire,

(b) obtain a relation for the variation of temperature in the wire by solving the differential equation, and

(c) determine the temperature at the centerline of the wire.

Assumptions

- The system is at steady-state

Givens

$r_o = 0.6\,\text{cm}$	Outer radius of wire
$k = 15.2\,\text{W m}^{-1}\,\text{K}^{-1}$	Thermal conductivity of wire
$\dot{e}_{\text{gen}} = 16.4\,\text{W cm}^{-3}$	Energy created in wire due to resistive heating
$T_\infty = 100\,^\circ\text{C}$	Temperature of water
$h = 3200\,\text{W m}^{-2}\,\text{K}^{-1}$	Heat transfer coefficient

Derivation

(a) The heat conduction equation on cylindrical coordinates is:

$$\frac{1}{r}\frac{\partial}{\partial r}\left(kr^2\frac{\partial T}{\partial r}\right)+\frac{1}{r^2\sin^2\theta}\frac{\partial}{\partial\phi}\left(k\frac{\partial T}{\partial\phi}\right)+\frac{1}{r^2\sin^2\theta}\frac{\partial}{\partial\theta}\left(k\sin\theta\frac{\partial T}{\partial\theta}\right)+\dot{e}_{\text{gen}}=\rho c\frac{\partial T}{\partial t}.$$

As we assume the wire is infinitely long $\frac{\partial T}{\partial z}=0$, as we assume the wire is homogeneous $\frac{\partial T}{\partial\phi}=0$ and as we assume steady state $\frac{\partial T}{\partial t}=0$. Hence the differential equation reduces to:

$$\frac{1}{r}\frac{\partial}{\partial r}\left(kr\frac{\partial T}{\partial r}\right)+\dot{e}_{\text{gen}}=0 \qquad\Longleftrightarrow\qquad \frac{1}{r}\frac{\partial}{\partial r}\left(k\frac{\partial T}{\partial r}\right)+\frac{\dot{e}_{\text{gen}}}{k}=0.$$

We need two boundary conditions to fully define this. Due to symmetry, $\left.\frac{\mathrm{d}T}{\mathrm{d}r}\right|_{r=0}=0$. Our second boundary condition comes from the specified conduction to the surrounding media, which corresponds to:

$$-k\left.\frac{\mathrm{d}T}{\mathrm{d}r}\right|_{r=r_o}=h\left(T(r_o)-T_\infty\right).$$

> **Result**
>
> Differential Equation:
>
> $$\frac{1}{r}\frac{\partial}{\partial r}\left(k\frac{\partial T}{\partial r}\right)+\frac{\dot{e}_{\text{gen}}}{k}=0.$$
>
> Boundary Condition 1:
>
> $$\left.\frac{\mathrm{d}T}{\mathrm{d}r}\right|_{r=0}=0.$$
>
> Boundary Condition 2:
>
> $$-k\left.\frac{\mathrm{d}T}{\mathrm{d}r}\right|_{r=r_o}=h\left(T(r_o)-T_\infty\right).$$

(b) We start with our differential equation:

$$\begin{aligned}\frac{1}{r}\frac{\mathrm{d}}{\mathrm{d}r}\left(r\frac{\mathrm{d}T}{\mathrm{d}r}\right)+\frac{\dot{e}_{\text{gen}}}{k}&=0\\ \frac{\mathrm{d}}{\mathrm{d}r}\left(r\frac{\mathrm{d}T}{\mathrm{d}r}\right)&=-\frac{\dot{e}_{\text{gen}}}{k}r\\ r\frac{\mathrm{d}T}{\mathrm{d}r}&=-\frac{\dot{e}_{\text{gen}}}{k}\frac{r^2}{2}+C_1.\end{aligned}$$

Our first boundary condition gives:

$$\left.\frac{\mathrm{d}T}{\mathrm{d}r}\right|_{r=0}=0 \quad\Longrightarrow\quad C_1=0 \quad\Longrightarrow\quad \frac{\mathrm{d}T}{\mathrm{d}r}=-\frac{\dot{e}_{\text{gen}}}{k}\frac{r}{2}.$$

A second integration yields:

$$T(r)=-\frac{\dot{e}_{\text{gen}}}{k}\frac{r^2}{4}+C_2.$$

Now, we apply the second boundary condition. At r_o:

$$-k\frac{\mathrm{d}T}{\mathrm{d}r}\bigg|_{r=r_o} = -k\left(-\frac{\dot{e}_{\text{gen}}}{k}\frac{r_o}{2}\right) = \frac{\dot{e}_{\text{gen}} r_o}{2}.$$

Hence,

$$\begin{aligned}
\frac{\dot{e}_{\text{gen}} r_o}{2} &= h\left(T(r_o) - T_\infty\right) \\
T(r_o) &= T_\infty + \frac{\dot{e}_{\text{gen}} r_o}{2h} \\
-\frac{\dot{e}_{\text{gen}}}{4k} r_o^2 + C_2 &= T_\infty + \frac{\dot{e}_{\text{gen}} r_o}{2h} \\
C_2 &= T_\infty + \frac{\dot{e}_{\text{gen}} r_o}{2h} + \frac{\dot{e}_{\text{gen}} r_o^2}{4k} \\
T(r) &= -\frac{\dot{e}_{\text{gen}}}{4k} r^2 + T_\infty + \frac{\dot{e}_{\text{gen}} r_o}{2h} + \frac{\dot{e}_{\text{gen}} r_o^2}{4k} \\
&= T_\infty + \frac{\dot{e}_{\text{gen}}}{2}\left(\frac{r_o}{h} + \frac{1}{2k}\left(r_o^2 - r^2\right)\right).
\end{aligned}$$

> **Result**
>
> The temperature distribution within the wire is given by:
>
> $$T(r) = T_\infty + \frac{\dot{e}_{\text{gen}}}{2}\left(\frac{r_o}{h} + \frac{1}{2k}\left(r_o^2 - r^2\right)\right).$$

(c) The temperature at the centerline is simply the temperature $T(0)$. Hence:

$$\begin{aligned}
T(0) &= T_\infty + \frac{\dot{e}_{\text{gen}}}{2}\left(\frac{r_o}{h} + \frac{1}{2k} r_o^2\right) \\
&= 100\,^\circ\text{C} + \frac{16.4\,\text{W cm}^{-3}}{2}\left(\frac{0.6\,\text{cm}}{3200\,\text{W m}^{-2}\,\text{K}^{-1}} + \frac{1}{2 \cdot 15.2\,\text{W m}^{-1}\,\text{K}^{-1}} \cdot (0.6\,\text{cm})^2\right) \\
&= 125\,^\circ\text{C}.
\end{aligned}$$

> **Result**
>
> Hence, the centerline temperature of the resistive wire is:
>
> $$T(0) = 125\,^\circ\text{C}.$$

Final result – Problem 5

(a) Differential Equation:

$$\frac{1}{r}\frac{\partial}{\partial r}\left(k\frac{\partial T}{\partial r}\right)+\frac{\dot{e}_{\text{gen}}}{k}=0.$$

Boundary Condition 1:

$$\left.\frac{\mathrm{d}T}{\mathrm{d}r}\right|_{r=0}=0.$$

Boundary Condition 2:

$$\left.-k\frac{\mathrm{d}T}{\mathrm{d}r}\right|_{r=r_o}=h\left(T(r_o)-T_\infty\right).$$

(b) The temperature distribution within the wire is given by:

$$T(r)=T_\infty+\frac{\dot{e}_{\text{gen}}}{2}\left(\frac{r_o}{h}+\frac{1}{2k}\left(r_o^2-r^2\right)\right).$$

(c) Hence, the centerline temperature of the resistive wire is:

$$T(0)=125\,^\circ\text{C}.$$

Problem 3–5

Derive the optimal radius for

$$\dot{Q} = \frac{T_1 - T_\infty}{R_{\text{ins}} + R_{\text{conv}}} = \frac{T_1 - T_\infty}{\frac{\ln(r_2/r_1)}{2\pi Lk} + \frac{1}{h(2\pi r_2 L)}}.$$

Assumptions

- The optimal radius is the one that maximizes heat transfer rate

Derivation

This is the critical radius of insulation problem from (Çengel and Ghajar 2015, Ch. 3–5). We note that the variable radius here is the radius of insulation, r_2. Hence, we are optimizing $\dot{Q}$ with respect to r_2 and therefore we apply derivation w.r.t. r_2 and then set $\mathrm{d}\dot{Q}/\mathrm{d}r_2 = 0$. We note that only r_2 is variable and everything else is constant under the added insulation. We set $T_1 - T_\infty = k_1, 2\pi Lk = k_2$, and $2h\pi L = k_3$ and differentiate as:

$$\begin{aligned}
\frac{\mathrm{d}\dot{Q}}{\mathrm{d}r_2} &= \frac{\mathrm{d}}{\mathrm{d}r_2} \frac{k_1}{\frac{\ln(\frac{r_2}{r_1})}{k_2} + \frac{1}{k_3 \cdot r_2}} \\
&= \frac{k_1 k_2 k_3 \left(k_2 - k_3 r_2\right)}{\left(k_3 r_2 \ln\left(\frac{r_2}{r_1}\right) + k_2\right)^2} \\
&= \frac{(T_1 - T_\infty) \cdot 2\pi Lk \cdot 2h\pi L \left(2\pi Lk - 2h\pi L r_2\right)}{\left(2h\pi L r_2 \ln\left(\frac{r_2}{r_1}\right) + 2\pi Lk\right)^2} \\
&= -\frac{2\pi hkL \left(T_1 - T_\infty\right)\left(hr_2 - k\right)}{\left(hr_2 \ln\left(\frac{r_2}{r_1}\right) + k\right)^2} \\
0 &= -\frac{2\pi hkL \left(T_1 - T_\infty\right)\left(hr_2 - k\right)}{\left(hr_2 \ln\left(\frac{r}{r_1}\right) + k\right)^2} \\
\implies r_2 &\to \frac{k}{h}.
\end{aligned}$$

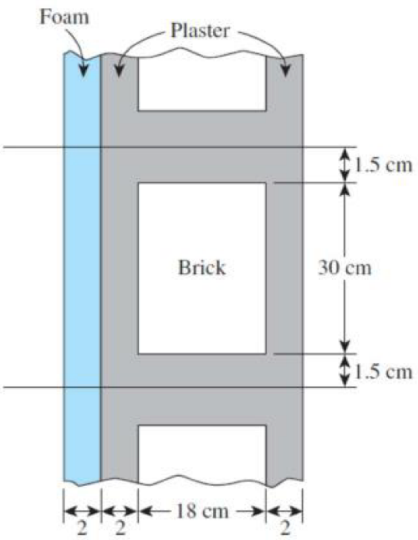
Final result – Problem 6

The optimal radius for insulation is therefore:

$$r_2 = \frac{k}{h}.$$

Problem 3–67

A 4 m high and 6 m wide wall consists of a long 18 cm × 30 cm cross section of horizontal bricks ($k = 0.72\,\mathrm{W\,m^{-1}\,K^{-1}}$) separated by 3 cm thick plaster layers ($k = 0.22\,\mathrm{W\,m^{-1}\,K^{-1}}$). There are also 2 cm thick plaster layers on each side of the wall, and a 2 cm thick rigid foam ($k = 0.026\,\mathrm{W\,m^{-1}\,K^{-1}}$) on the inner side of the wall. The indoor and the outdoor temperatures are 22 °C and −4 °C, and the convection heat transfer coefficients on the inner and the outer sies are $h_1 = 10\,\mathrm{W\,m^{-2}\,K^{-1}}$ and $h_2 = 20\,\mathrm{W\,m^{-2}\,K^{-1}}$, respectively. Assuming one-dimensional heat transfer and disregarding radiation, determine the rate of heat transfer through the wall.



Assumptions

- The system is at steady state
- There are no contact resistances

Givens

$h_{\mathrm{wall}} = 4\,\mathrm{m}$	Height of wall
$w_{\mathrm{wall}} = 6\,\mathrm{m}$	Width of wall
$w_{\mathrm{brick}} = 18\,\mathrm{cm}$	Width of brick
$h_{\mathrm{brick}} = 30\,\mathrm{cm}$	Height of brick
$k_{\mathrm{brick}} = 0.72\,\mathrm{W\,m^{-1}\,K^{-1}}$	Thermal conductivity of brick
$h_{\mathrm{plaster}} = 3\,\mathrm{cm}$	Height of plaster layer between bricks
$w_{\mathrm{plaster}} = 4\,\mathrm{cm}$	Total width of plaster on both sides of the bricks
$k_{\mathrm{plaster}} = 0.22\,\mathrm{W\,m^{-1}\,K^{-1}}$	Thermal conductivity of plaster
$w_{\mathrm{foam}} = 2\,\mathrm{cm}$	Width of foam
$k_{\mathrm{foam}} = 0.026\,\mathrm{W\,m^{-1}\,K^{-1}}$	Thermal conductivity of foam
$T_{\mathrm{in}} = 22\,^{\circ}\mathrm{C}$	Indoor temperature
$T_{\mathrm{out}} = -4\,^{\circ}\mathrm{C}$	Outdoor temperature
$h_1 = 10\,\mathrm{W\,m^{-2}\,K^{-1}}$	Inner heat convection heat transfer coefficient
$h_2 = 20\,\mathrm{W\,m^{-2}\,K^{-1}}$	Outer heat convection heat transfer coefficient.

Derivation

The heat transfer rate is simply the quotient of the temperature difference over the interface $\Delta T_{\text{interface}} = T_{\text{in}} - T_{\text{out}} = 26\,\text{K}$ and the thermal resistance R_{wall}:

$$\dot{Q} = \frac{\Delta T_{\text{interface}}}{R_{\text{wall}}}.$$

We only consider the area around one brick and its associated plaster. The thermal resistance network kan then be seen as a series connection of 2 cm foam, 2×2 cm plaster which is then connected to a 18 cm parallel connection of 30 cm height of brick and 3 cm height of plaster.

The middle parallel section has area fraction of bricks of

$$f_{\text{brick}} = \frac{30}{33}$$

and of plaster of

$$f_{\text{plaster}} = \frac{3}{33}.$$

The total wall area, A, is

$$A = 4\,\text{m} \cdot 6\,\text{m} = 24\,\text{m}^2.$$

We are now able to replace the middle parallel section with an equivalent thermal conductivity

$$k_{\text{eq}} = f_{\text{brick}} k_{\text{brick}} + f_{\text{plaster}} k_{\text{plaster}} = \frac{30}{33} \cdot 0.72\,\text{W m}^{-1}\,\text{K}^{-1} + \frac{3}{33} 0.22\,\text{W m}^{-1}\,\text{K}^{-1} = 0.6745\,\text{W m}^{-1}\,\text{K}^{-1}.$$

Now we can calculate the value area-normalized value of R to avoid having to multiply everything by the area A thorughout, as:

$$\begin{aligned} R &= \frac{1}{h_1} + \frac{w_{\text{foam}}}{k_{\text{foam}}} + 2 \cdot \frac{w_{\text{plaster}}}{k_{\text{plaster}}} + \frac{w_{\text{brick}}}{k_{eq}} + \frac{1}{h_2} \\ &= \frac{1}{10\,\text{W m}^{-2}\,\text{K}^{-1}} + \frac{2\,\text{cm}}{0.026\,\text{W m}^{-1}\,\text{K}^{-1}} + 2 \cdot \frac{2\,\text{cm}}{0.22\,\text{W m}^{-1}\,\text{K}^{-1}} + \frac{18\,\text{cm}}{0.6745\,\text{W m}^{-1}\,\text{K}^{-1}} + \frac{1}{20\,\text{W m}^{-2}\,\text{K}^{-1}} \\ &= 1.368\,\text{m}^2\,\text{K W}^{-1}. \end{aligned}$$

Hence, the total heat transfer rate is:

$$\dot{Q} = \frac{A \Delta T}{R} = \frac{24\,\text{m}^2 \cdot 26\,\text{K}}{1.368\,\text{m}^2\,\text{K W}^{-1}} = 456\,\text{W}.$$

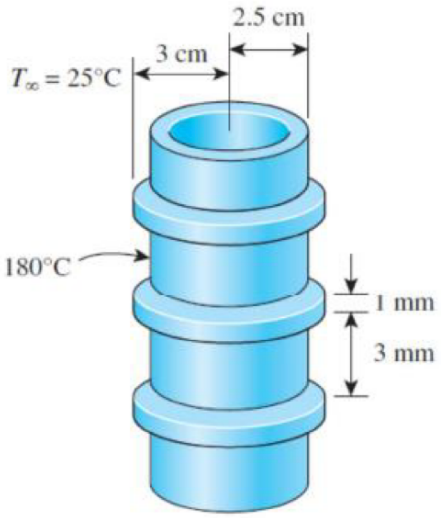
Final result – Problem 7

The total heat transfer rate over the entire wall is:

$$\dot{Q} = 456\,\text{W}.$$

Problem 3–128

Steam in a heating system flows through tubes whose outer diameter is 5 cm and whose walls are maintained at a temperature of 180 °C. Circular aluminum alloy 2024–T6 fins ($k = 186\,\mathrm{W\,m^{-1}\,K^{-1}}$) of outer diameter 6 cm and constant thickness 1 mm are attached to the tube. The space between the fins is 3 mm, and thus there are 250 fins per meter length of the tube. Heat is transferred to the surrounding air at $T_\infty = 25\,^\circ\mathrm{C}$, with a heat transfer coefficient of $40\,\mathrm{W\,m^{-2}\,K^{-1}}$. Determine the increase in heat transfer from the tube per meter of its length as a result of adding fins.



Givens

$r_{\text{tube, outer}}$	2.5 cm
$t = 1\,\mathrm{mm}$	Thickness of fins
$T_{\text{wall}} = 180\,^\circ\mathrm{C}$	Temperature of the tube walls
$k_{\text{fins}} = 186\,\mathrm{W\,m^{-1}\,K^{-1}}$	Thermal conductivity of fins
$r_{\text{fin, outer}} = 3\,\mathrm{cm}$	outer diameter of fin
$N = 250$	Amount of fins per meter
$T_\infty = 25\,^\circ\mathrm{C}$	Temperature of surrounding air
$h = 40\,\mathrm{W\,m^{-2}\,K^{-1}}$	Heat transfer coefficient

Derivation

Here, we are looking to first determine the fin effectiveness ϵ_{fin}:

$$\epsilon_{\text{fin}} = \frac{\dot{Q}_{\text{fin}}}{\dot{Q}_{\text{no fin}}} = \frac{\dot{Q}_{\text{fin}}}{hA_b\theta_b}.$$

The base area covered by one fin is:

$$A_b = 2\pi r_{\text{tube, outer}} t = 2\pi \cdot 2.5\,\mathrm{cm} \cdot 1\,\mathrm{mm} = 1.5708\,\mathrm{cm^2}.$$

So without any fins, this little piece of pipe would transfer heat at a rate of

$$\dot{Q}_{\text{no fin}} = hA_b\theta_b = 40\,\mathrm{W\,m^{-2}\,K^{-1}} \cdot 1.571\,\mathrm{cm^2} \cdot 155\,\mathrm{K} = 0.974\,\mathrm{W}.$$

Now, we need to determine the heat transfer from a single fin $\dot{Q}_{\text{fin}}$. This is given as:

$$\dot{Q}_{\text{fin}} = \eta_{\text{fin}} A_{\text{fin}} h \theta_b.$$

From the course book we know that for circular fins of rectangular profile the fin parameter m, the corrected outer radius r_{2c}, the fin area A_{fin}, the fin efficiency η_{fin}, and the constant C_2 are respectively (Çengel and Ghajar 2015, p. 200):

$$\begin{aligned}
m &= \sqrt{\frac{2h}{kt}} \\
r_{2c} &= r_2 + \frac{t}{2} \\
\eta_{\text{fin}} &= C_2 \frac{K_1(mr_1)I_1(mr_{2c}) - I_1(mr_1)K_1(mr_{2c})}{I_0(mr_1)K_1(mr_{2c}) + K_0(mr_1)I_1(mr_{2c})} \\
C_2 &= \frac{\frac{2r_1}{m}}{r_{2c}^2 - r_1^2}.
\end{aligned}$$

We start by finding the fin parameter, the corrected outer radius, the fin area, and the constant C_2 as:

$$\begin{aligned}
m &= \sqrt{\frac{2 \cdot 40\,\text{W m}^{-2}\,\text{K}^{-1}}{186\,\text{W m}^{-1}\,\text{K}^{-1} \cdot 1\,\text{mm}}} = 20.739\,\text{m}^{-1} \\
r_{2c} &= 3\,\text{cm} + \frac{1\,\text{mm}}{2} = 30.5\,\text{mm} \\
A_{\text{fin}} &= 2\pi \cdot \left((30.5\,\text{mm})^2 - (2.5\,\text{cm})^2\right) = 1917.94\,\text{mm}^2 \\
C_2 &= \frac{\frac{2 \cdot 2.5\,\text{cm}}{20.739\,\text{m}^{-1}}}{(30.5\,\text{mm})^2 - (2.5\,\text{cm})^2} = 7.89.
\end{aligned}$$

The parameters I_1, I_0, K_1, and K_0 are the modified Bessel functions. The arguments to these are:

$$\begin{aligned}
mr_1 &= 20.739\,\text{m}^{-1} \cdot 2.5\,\text{cm} = 0.5185 \\
mr_{2c} &= 20.739\,\text{m}^{-1} \cdot 30.5\,\text{mm} = 0.6325.
\end{aligned}$$

The modified Bessel functions can then be evaluated at these points using the Wolfram engine, which allows is to calculate the fin efficiency:

$$\eta_{\text{fin}} = 7.89 \cdot \frac{K_1(0.5185)I_1(0.6325) - I_1(0.5185)K_1(0.6325)}{I_0(0.5185)K_1(0.6325) + K_0(0.5185)I_1(0.6325)} = 0.993\,586.$$

Hence, the heat transfer from a single fin is:

$$\dot{Q}_{\text{fin}} = \eta_{\text{fin}} A_{\text{fin}} h \theta_b = 0.993\,586 \cdot 1917.94\,\text{mm}^2 \cdot 40\,\text{W m}^{-2}\,\text{K}^{-1} \cdot 155\,\text{K} = 11.815\,\text{W}.$$

Hence, the fin effectiveness is:

$$\epsilon_{\text{fin}} = \frac{\dot{Q}_{\text{fin}}}{\dot{Q}_{\text{no fin}}} = \frac{11.815\,\text{W}}{0.974\,\text{W}} = 12.130.$$

The increase in heat transfer caused by a single fin is:

$$\dot{Q}_{one\,fin} = \dot{Q}_{\text{fin}} - \dot{Q}_{\text{no fin}} = 11.815\,\text{W} - 0.974\,\text{W} = 10.841\,\text{W}.$$

There are $N = 250\,\text{m}^{-1}$ fins so:

$$\Delta\dot{Q}_{\text{per meter}} = N \cdot \dot{Q}_{\text{one fin}} = 250\,\text{m}^{-1} \cdot 10.841\,\text{W} = 2.710\,\text{kW m}^{-1}.$$

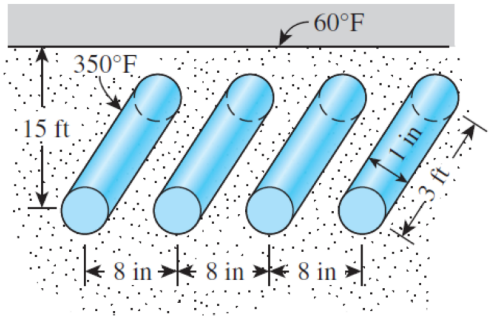
Final result – Problem 8

The increase in heat transfer rate per meter as a result of adding fins is thus

$$\Delta\dot{Q}_{\text{per meter}} = 2.710\,\text{kW m}^{-1}.$$

Problem 3–146E

A row of 1 m long and 2.5 cm diameter used uranium fuel rods that are still radioactive are buried in the ground parallel to each other with a center-to-center distance of 20 cm at a depth of 4.5 m from the ground surface at a location where the thermal conductivity of the soil is $1.1\,\mathrm{W\,m^{-1}\,K^{-1}}$. If the surface temperature of the rods and the ground are 175 °C and 15 °C respectively, determine the rate of heat transfer from the fuel rods to the atmosphere through the soil.



Assumptions

- The system is at steady state

Givens

$L = 1\,\mathrm{m}$	Length of rods
$D = 2.5\,\mathrm{cm}$	Diameter of rods
$w = 20\,\mathrm{cm}$	Center-to-center distance between rods
$z = 4.5\,\mathrm{m}$	Depth of rods
$k = 1.1\,\mathrm{W\,m^{-1}\,K^{-1}}$	Thermal conductivity of the soil
$T_1 = 175\,^\circ\mathrm{C}$	Temperature of rods
$T_2 = 15\,^\circ\mathrm{C}$	Temperature of ground
$N = 4$	Amount of rods

Derivation

Here we need to use the conduction shape factor S to find the rate of heat transfer using (Çengel and Ghajar 2015, Eq. 3–93):

$$\dot{Q} = Sk(T_1 - T_2)\,.$$

For a row of equally spaced parallel isothermal cylinders buried in a semi-infinite medium, where $L \gg D, z$, and $w > 1.5D$ (as in this case) the conduction shape factor is given by (Çengel and Ghajar 2015, Table 3–7(4)):

$$S = \frac{2\pi L}{\ln\left(\frac{2w}{\pi D}\sinh\frac{2\pi z}{w}\right)} = \frac{2\cdot\pi\cdot 1\,\mathrm{m}}{\ln\left(\frac{2\cdot 20\,\mathrm{cm}}{\pi\cdot 2.5\,\mathrm{cm}}\cdot\sinh\left(\frac{2\cdot\pi\cdot 4.5\,\mathrm{m}}{20\,\mathrm{cm}}\right)\right)} = 4.4153\,\mathrm{cm}$$

per cylinder. Hence,

$$\dot{Q}_{\text{per cylinder}} = 4.4153\,\text{cm} \cdot 1.1\,\text{W m}^{-1}\,\text{K}^{-1} \cdot (175\,^\circ\text{C} - 15\,^\circ\text{C}) = 7.7709\,\text{W}.$$

So the total rate of heat transfer is:

$$\dot{Q}_{\text{total}} = \dot{Q}_{\text{per cylinder}} \cdot N = 7.7709\,\text{W} \cdot 4 = 31.0836\,\text{W}.$$

Final result – Problem 9

The total rate of heat transfer is therefore

$$\dot{Q}_{\text{total}} = 31.084\,\text{W}.$$

Lecture 4: Conduction, Pt. III 2026-09-15

Problem 4–17

Metal plates ($k = 180\,\mathrm{W\,m^{-1}\,K^{-1}}$, $\rho = 2800\,\mathrm{kg\,m^{-3}}$, and $c_p = 880\,\mathrm{J\,kg^{-1}\,K^{-1}}$) with a thickness of 1 cm are being heated in an oven for 2 minutes. Air in the oven is maintained at 800 °C with a convection heat transfer coefficient of $200\,\mathrm{W\,m^{-2}\,K^{-1}}$. If the initial temperature of the plates is 20 °C, determine the temperature of the plates when they are removed from the oven.

Assumptions

- The metal plates are lumped

Givens

$k = 180\,\mathrm{W\,m^{-1}\,K^{-1}}$	Thermal conductivity of metal plates
$\rho = 2800\,\mathrm{kg\,m^{-3}}$	Density of metal plates
$c_p = 880\,\mathrm{J\,kg^{-1}\,K^{-1}}$	Heat capacity of metal plates
$z = 1\,\mathrm{cm}$	Thickness of metal plates
$t = 2\,\mathrm{min}$	Heating time
$T_\infty = 800\,^\circ\mathrm{C}$	Temperature of oven
$h = 200\,\mathrm{W\,m^{-2}\,K^{-1}}$	Convection heat transfer coefficient
$T_i = 20\,^\circ\mathrm{C}$	Initial temperature of metal plates

Derivation

From (Çengel and Ghajar 2015, Eqs. 4–4 & 4–5) the temperature $T(t)$ is:

$$T(t) = T_\infty + (T_i - T_\infty)\, e^{-bt}$$

where

$$b = \frac{hA_s}{\rho \mathscr{V} c_p} = \frac{h}{\rho \cdot L_c \cdot c_p}$$

is the reciprocal of the time constant.

The characteristic length L_c for a thin plate heated from both sides will be $L_c = z/2 = 0.5\,\mathrm{cm}$ The reciprocal of the time constant is determined to be:

$$b = \frac{200\,\mathrm{W\,m^{-2}\,K^{-1}}}{2800\,\mathrm{kg\,m^{-3}} \cdot 0.5\,\mathrm{cm} \cdot 880\,\mathrm{J\,kg^{-1}\,K^{-1}}} = 0.016\,23\,\mathrm{Hz}.$$

Hence the temperature after $t = 2\,\text{min}$ is:

$$T(2\,\text{min}) = 800\,^\circ\text{C} + (20\,^\circ\text{C} - 800\,^\circ\text{C}) \cdot e^{-0.016\,23\,\text{Hz}\cdot 2\,\text{min}} = 688.8\,^\circ\text{C}.$$

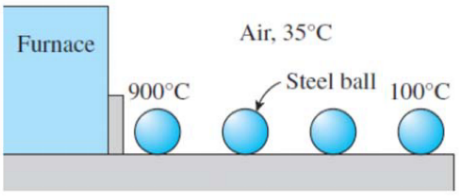
Final result – Problem 10

Hence, the temperaure of the metal plates, when they are removed from the oven after $2\,\text{min}$ is

$$T(2\,\text{min}) = 688.8\,^\circ\text{C}.$$

Problem 4–34

Carbon steel balls ($\rho = 7833\,\mathrm{kg\,m^{-3}}, k = 54\,\mathrm{W\,m^{-1}\,K^{-1}}, c_p = 0.465\,\mathrm{kJ\,kg^{-1}\,{}^\circ C^{-1}}$, and $\alpha = 1.474 \cdot 10^{-6}\,\mathrm{m^2\,s^{-1}}$) 8 mm in diameter are annealed by heating them first to 900 °C in a furnace and then allowing them to cool slowly to 100 °C in ambient air at 35 °C. If the average heat transfer coefficient is $75\,\mathrm{W\,m^{-2}\,K^{-1}}$, determine how long the annealing process will take. If 2500 balls are to be annealed per hour, determine the total rate of heat transfer from the balls to the ambient air.



Givens

$\rho = 7833\,\mathrm{kg\,m^{-3}}$	Density of carbon steel
$k = 54\,\mathrm{W\,m^{-1}\,K^{-1}}$	Thermal conductivity of carbon steel
$c_p = 0.465\,\mathrm{kJ\,kg^{-1}\,{}^\circ C^{-1}}$	Heat capacity of carbon steel
$\alpha = 1.474 \cdot 10^{-6}\,\mathrm{m^2\,s^{-1}}$	Thermal diffusivity of carbon steel
$d = 8\,\mathrm{mm}$	Diameter of steel balls
$T_1 = 900\,^\circ\mathrm{C}$	Temperature before cooling
$T_2 = 100\,^\circ\mathrm{C}$	Temperature after cooling
$T_\infty = 35\,^\circ\mathrm{C}$	Ambient temperature
$h = 75\,\mathrm{W\,m^{-2}\,K^{-1}}$	Heat transfer coefficient
$n = 2500\,\mathrm{h^{-1}}$	Rate of balls to be annealed

Derivation

We start by determining whether the steel balls can be viewed as lumped. For a sphere of radius r_o, the characteristic length is $L_c = r_o/3$. Hence,

$$L_c = \frac{r_o}{3} = \frac{d}{6} = \frac{8\,\mathrm{mm}}{6} = 1.333\,\mathrm{mm}.$$

The Biot number can then be calculated to be:

$$\mathrm{Bi} = \frac{hL_c}{k} = \frac{75\,\mathrm{W\,m^{-2}\,K^{-1}} \cdot 1.333\,\mathrm{mm}}{54\,\mathrm{W\,m^{-1}\,K^{-1}}} = 0.001\,85.$$

As $\mathrm{Bi} = 0.0019 < 0.1$ the system can be viewed as being lumped.

We once again employ:

$$T(t) = T_\infty + (T_i - T_\infty)\, e^{-bt}$$

and solve for the required time to obtain:

$$t = \frac{\ln\left(\frac{T_2 - T_\infty}{(T_1 - T_\infty)}\right)}{b}.$$

The reciprocal of the time constant is:

$$b = \frac{h}{\rho L_c c_p} = \frac{75\,\mathrm{W\,m^{-2}\,K^{-1}}}{7833\,\mathrm{kg\,m^{-3}} \cdot 1.333\,\mathrm{mm} \cdot 0.465\,\mathrm{kJ\,kg^{-1}\,{}^\circ C^{-1}}} = 0.015\,45\,\mathrm{Hz}.$$

Hence, the time required to cool down the balls during the annealing is:

$$t = -\frac{\ln\left(\frac{100\,^\circ\mathrm{C} - 35\,^\circ\mathrm{C}}{900\,^\circ\mathrm{C} - 35\,^\circ\mathrm{C}}\right)}{0.015\,45\,\mathrm{Hz}} = 2.79\,\mathrm{Minutes} \rightarrow 167.4\,\mathrm{s}.$$

The total rate of heat transfer over the time interval $t = 0$ to t is:

$$\dot{Q} = \dot{m} c_p \left(T(t) - T_i\right) \implies \dot{Q} = N \cdot \rho \cdot \frac{4}{3}\pi \cdot \left(\frac{d}{2}\right)^3 \cdot c_p \left(T_2 - T_1\right).$$

Hence,

$$\dot{Q} = 2500\,\mathrm{h^{-1}} \cdot 7833\,\mathrm{kg\,m^{-3}} \cdot \frac{4}{3} \cdot \pi \cdot \left(\frac{8\,\mathrm{mm}}{2}\right)^3 \cdot 0.465\,\mathrm{kJ\,kg^{-1}\,{}^\circ C^{-1}} \cdot (100\,^\circ\mathrm{C} - 900\,^\circ\mathrm{C}) = -0.5425\,\mathrm{kW}.$$

Final result – Problem 11

The cooling part of the annealing process will take:

$$t = 167.4\,\mathrm{s}$$

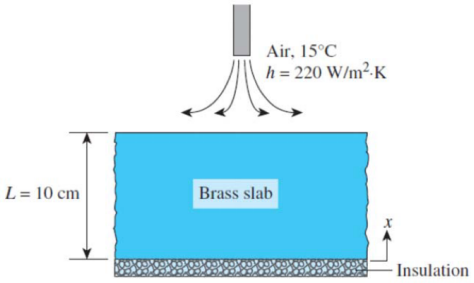
not enough information is known about the heating part to estimate the duration of this part. The $2500\,\mathrm{h^{-1}}$ balls will give of heat at a rate of

$$\dot{Q} = 0.5425\,\mathrm{kW}$$

to the surroundings.

Problem 4–47

A hot brass plate is having its upper surface cooled by impinging jet of air at temperature of 15 °C and convection heat transfer coefficient of 220 $\mathrm{W\,m^{-2}\,K^{-1}}$. The 10 cm thick brass plate ($\rho = 8530\,\mathrm{kg\,m^{-3}}, c_p = 380\,\mathrm{J\,kg^{-1}\,K^{-1}}, k = 110\,\mathrm{W\,m^{-1}\,K^{-1}}$, and $\alpha = 33.9 \cdot 10^{-6}\,\mathrm{m^2\,s^{-1}}$ has a uniform initial temperature of 650 °C, and the bottom surface of the plate is insulated. Determine the temperature at the center plane of the brass plate after 3 minutes of cooling. Solve this problem using analytical one-term approximation (not the Heisler charts).



Assumptions

- No heat is lost through the insulated bottom portion

Givens

$T_\infty = 15\,^\circ\mathrm{C}$	Temperature of air jet
$h = 220\,\mathrm{W\,m^{-2}\,K^{-1}}$	Convection heat transfer coefficient
$z = 10\,\mathrm{cm}$	Thickness of brass plate
$\rho = 8530\,\mathrm{kg\,m^{-3}}$	Density of brass
$c_p = 380\,\mathrm{J\,kg^{-1}\,K^{-1}}$	Heat capacity of brass
$k = 110\,\mathrm{W\,m^{-1}\,K^{-1}}$	Thermal conductivity of brass
$\alpha = 33.9 \cdot 10^{-6}\,\mathrm{m^2\,s^{-1}}$	Thermal diffusivity of brass
$T_i = 650\,^\circ\mathrm{C}$	Initial temperature
$t = 3\,\mathrm{min}$	Time

Derivation

For good measure, we start by determining whether the system can be viewed as lumped as above. In this case the characteristic length is $L_c = z = 10\,\mathrm{cm}$ as no heating occurs from underneath. Hence

$$\mathrm{Bi} = \frac{hL_c}{k} = \frac{220\,\mathrm{W\,m^{-2}\,K^{-1}} \cdot 10\,\mathrm{cm}}{110\,\mathrm{W\,m^{-1}\,K^{-1}}} = 0.2.$$

As $\mathrm{Bi} = 0.2 \not\leq 0.1$ the system can not be regarded as being lumped.

The analytical one term solution for a plane wall is (in that we regard it as a symmetrically bounded wall with twice the thickness):

$$\theta = A_1 e^{-\lambda_1^2 \mathrm{Fo}} \cdot \cos\left(\lambda_1 \frac{x}{L}\right).$$

The values $\lambda_1(\mathrm{Bi})$ and $A_1(\mathrm{Bi})$ are found in (Çengel and Ghajar 2015, Table 4–2) for a plane wall to be $\lambda_1(0.2) = 0.4328$ and $A_1(0.2) = 1.0311$. The Fourier number Fo is defined as:

$$\mathrm{Fo} = \frac{\alpha t}{L^2} \implies \mathrm{Fo} = \frac{33.9 \cdot 10^{-6}\,\mathrm{m^2\,s^{-1}} \cdot 3\,\mathrm{min}}{(10\,\mathrm{cm})^2} = 0.6102.$$

The location we are requested to find the temperature at is the midline $x = z/2 = 5\,\mathrm{cm}$. Hence,

$$\theta = 1.0311 \cdot e^{-0.4328^2 \cdot 0.6102} \cdot \cos\left(0.4328 \cdot \frac{5\,\mathrm{cm}}{10\,\mathrm{cm}}\right) = 0.898.$$

Hence, the temperature excess $\theta = 0.898$. This is defined as:

$$\theta = \frac{T - T_\infty}{T_i - T_\infty}.$$

Hence, the temperature at the center is:

$$T = T_\infty + (T_i - T_\infty) \cdot 0.898 = 15\,^\circ\mathrm{C} + (650\,^\circ\mathrm{C} - 15\,^\circ\mathrm{C}) \cdot 0.898 = 555.23\,^\circ\mathrm{C}.$$

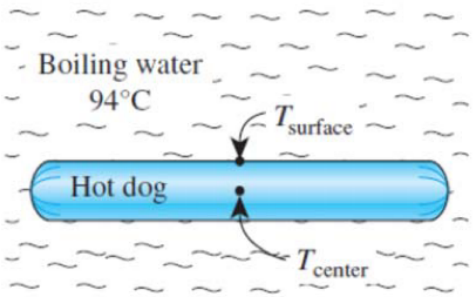
Final result – Problem 12

The temperature at the center of the brass after $t = 3\,\mathrm{min}$ is

$$T = 555\,^\circ\mathrm{C}.$$

Problem 4-66

A father and son conducted the following simple experiment on a hot dog which measured 12.5 cm in length and 2.2 cm in diameter. They inserted one food thermometer into the midpoint of the hot dog and another one was placed just under the skin of the hot dog. The temperature of the thermometers were monitored until both temperatures read 20 °C which is the ambient temperature. The hot dog was then placed in 94 °C boiling water and after exactly 2 minutes they recorded the center temperature and the skin temperature of the hot dog to be 59 °C and 88 °C, respectively. Assuming the following properties for the hot dog: $\rho = 980\,\mathrm{kg\,m^{-3}}$ and $c_p = 3900\,\mathrm{J\,kg^{-1}\,K^{-1}}$ and using transient temperature charts, determine



(a) the thermal diffusivity of the hot dog,

(b) the thermal conductivity of the hot dog, and

(c) the convection heat transfer coefficient.

Assumptions

- The properties of the hotdog are totally uniform throughout
- The hotdog is cylindrical

Givens

$l = 12.5\,\mathrm{cm}$	Length of hotdog
$d = 2.2\,\mathrm{cm}$	Diameter of hotdog
$T_i = 20\,^\circ\mathrm{C}$	Air temperature and initial temperature of hotdog
$T_\infty = 94\,^\circ\mathrm{C}$	Temperature of water
$T_s = 89\,^\circ\mathrm{C}$	Skin temperature of hotdog after heating
$T_{\mathrm{center}} = 59\,^\circ\mathrm{C}$	Center temperature of hotdog after heating
$t = 2\,\mathrm{min}$	Duration of heating
$\rho = 980\,\mathrm{kg\,m^{-3}}$	Density of hotdog
$c_p = 3900\,\mathrm{J\,kg^{-1}\,K^{-1}}$	Heat capacity of hotdog

Derivation

(a) We start by determining the dimensionless temperature at the surface of the hotdog and at its center respectively as:

$$\theta_s = \frac{T_s - T_\infty}{T_i - T_\infty} = \frac{89\,^\circ\mathrm{C} - 94\,^\circ\mathrm{C}}{20\,^\circ\mathrm{C} - 94\,^\circ\mathrm{C}} = 0.067\,57$$
$$\theta_0 = \frac{T_{\text{center}} - T_\infty}{T_i - T_\infty} = \frac{59\,^\circ\mathrm{C} - 94\,^\circ\mathrm{C}}{20\,^\circ\mathrm{C} - 94\,^\circ\mathrm{C}} = 0.472\,97.$$

The ratio between these is:

$$\frac{\theta_s}{\theta_0} = \frac{0.067\,57}{0.472\,97} = 0.142\,863.$$

At the surface $r/r_o = 1$, hence we find the intersection between the 1.0 curve and $\theta/\theta_0 = 0.143$ on (Çengel and Ghajar 2015, Fig. 4–18(b)). Here, we see that $\mathrm{Bi}^{-1} = 0.14 \implies \mathrm{Bi} \approx 7$. Now, we use (Çengel and Ghajar 2015, Fig 4–18(a)) where we use $\mathrm{Bi}^{-1} = 0.14$ and $\theta_s/\theta_0 = 0.1428$ and find $\mathrm{Fo} = 0.45$. Hence:

$$\begin{aligned}
\mathrm{Fo} &= \alpha\frac{t}{r_o^2} \\
\alpha &= \frac{\mathrm{Fo}\cdot r_o^2}{t} \\
&= \frac{0.45\cdot(1.1\,\mathrm{cm})^2}{2\,\mathrm{min}} \\
&= 0.272\,25\,\mathrm{cm}^2/\mathrm{min} = 4.5375\cdot 10^{-7}\,\mathrm{m}^2\,\mathrm{s}^{-1}.
\end{aligned}$$

> **Result**
>
> The thermal diffusivity of the hotdog is approximately
>
> $$\alpha = 4.54\cdot 10^{-7}\,\mathrm{m}^2\,\mathrm{s}^{-1}.$$

(b) The thermal diffusivity is related to the thermal conductivity as:

$$\alpha = \frac{k}{\rho c_p} \implies k = \alpha\rho c_p.$$

Hence,

$$k = 4.5375\cdot 10^{-7}\,\mathrm{m}^2\,\mathrm{s}^{-1}\cdot 980\,\mathrm{kg\,m}^{-3}\cdot 3900\,\mathrm{J\,kg}^{-1}\,\mathrm{K}^{-1} = 1.734\,\mathrm{W\,K}^{-1}\,\mathrm{m}^{-1}.$$

> **Result**
>
> The thermal conductivity of the hotdog is
>
> $$k = 1.734\,\mathrm{W\,K}^{-1}\,\mathrm{m}^{-1}.$$

(c) The convection heat transfer coefficient is related to the Biot number and thermal conductivity through:

$$\mathrm{Bi} = \frac{hL_c}{k} \implies h = \frac{\mathrm{Bi}\cdot k}{L_c}.$$

Hence,

$$h = \frac{7 \cdot 1.734\,\mathrm{W\,K^{-1}\,m^{-1}}}{\frac{1.1\,\mathrm{cm}}{2}} = 0.22\,\mathrm{W\,cm^{-2}\,K^{-1}} \to 2200\,\mathrm{W\,m^{-2}\,K^{-1}}.$$

Result

The conduction heat transfer coefficient is:

$$h = 2200\,\mathrm{W\,m^{-2}\,K^{-1}}.$$

Final result – Problem 13

(a) The thermal diffusivity of the hotdog is approximately

$$\alpha = 4.54 \cdot 10^{-7}\,\mathrm{m^2\,s^{-1}}.$$

(b) The thermal conductivity of the hotdog is

$$k = 1.734\,\mathrm{W\,K^{-1}\,m^{-1}}.$$

(c) The conduction heat transfer coefficient is:

$$h = 2200\,\mathrm{W\,m^{-2}\,K^{-1}}.$$

References

Çengel, Yunus A. and Afshin J. Ghajar (2015). *Heat and Mass Transfer: Fundamentals & Applications*. 5th edition. New York, NY: McGraw-Hill Education. 968 pp. ISBN: 978-0-07-339818-1.